PGR106
Linear Algebra

Lecture 7 – Vector Subspaces & Orthogonality

Kristiania University College

Vector Subspaces

- Vectors are elements of sets called vector spaces.
- A vector space is a collection of vectors, which can be added together and multiplied by scalars (numbers), satisfying specific rules.
- For the vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ which belong to vector space $\boldsymbol{V}$
  - Vector addition of $\boldsymbol{x}$ and $\boldsymbol{y}$ also belongs to the vector space $\boldsymbol{V}$
  - Scalar multiplication of $\boldsymbol{c}$ with $\boldsymbol{x}$ also belongs to the vector space $\boldsymbol{V}$
  - There exists a zero vector which also belongs to the vector space $\boldsymbol{V}$

- Any linear combination of vectors in a vector space $\boldsymbol{V}$, also belongs to $\boldsymbol{V}$.

Vector Subspaces

- Subspace of a vector space V is a subset of V.
- R^n is a vector space consisting of all the **n** dimensional vectors.
- R^3 is a 3-dimensional space and the subspace of R^3 is a plane.
- Subspaces give useful information about matrices.
  - Null space (Kernel)
  - Column space (Range)

Null Space

- Null space of a matrix ***A*** (***Nul(A)***) consists of all solutions ***x*** to the equation Ax = 0.

$$3x_1 - x_2 = 0$$

$$-6x_1 + 2x_2 = 0$$

$$A = \begin{bmatrix} 3 & -1 \\ -6 & 2 \end{bmatrix}$$

$$Ax = 0$$

$$\left[\begin{array}{cc|c} 3 & -1 & 0 \\ -6 & 2 & 0 \end{array}\right]$$

Augmented Matrix

Null Space

Augmented Matrix

$$\left[\begin{array}{cc|c} 3 & -1 & 0 \\ -6 & 2 & 0 \end{array}\right]$$

$R_2 \leftarrow R_2 + 2R_1$

$$\left[\begin{array}{cc|c} 3 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$R_1 \leftarrow \frac{R_1}{3}$

$$\left[\begin{array}{cc|c} 1 & -1/3 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

Null Space

Augmented Matrix

$$\left[\begin{array}{cc|c} 3 & -1 & 0 \\ -6 & 2 & 0 \end{array}\right] \xrightarrow{\text{Reduced Row Echelon Form}} \begin{array}{ccc} x_1 & x_2 & \\ \left[\begin{array}{c} 1 \\ 0 \end{array}\right. & \begin{array}{c} -1/3 \\ 0 \end{array} & \left|\begin{array}{c} 0 \\ 0 \end{array}\right] \end{array}$$

$$x_1 - \frac{1}{3}x_2 = 0$$

$$x_1 = \frac{1}{3}x_2$$

Null Space

$$3x_1 - x_2 = 0$$

$$-6x_1 + 2x_2 = 0$$

$$A = \begin{bmatrix} 3 & -1 \\ -6 & 2 \end{bmatrix}$$

$$\boxed{x_1 = \frac{1}{3}x_2}$$

$x_1 = 1$

$x_2 = 3$

$$x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$x_1 = 2$

$x_2 = 6$

$$x = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

$x_1 = 0$

$x_2 = 0$

$$x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Null Space

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 2 \\ -7 & 5 & -1 \end{bmatrix} \qquad u = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$\boxed{Au = 0}$$

$$\begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 2 \\ -7 & 5 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5 * (-1) + (-1) * (-1) + 2 * 2 \\ (-2) * (-1) + 6 * (-1) + 2 * 2 \\ (-7) * (-1) + 5 * (-1) + (-1) * 2 \end{bmatrix} = \begin{bmatrix} -5 + 1 + 4 \\ 2 - 6 + 4 \\ 7 - 5 - 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Null Space & Linear Independence

- Null space of a matrix ***A*** (***Nul(A)***) consists of all solutions ***x*** to the equation $Ax = 0$.

$$v_1 = \begin{bmatrix} 5 \\ -2 \\ -7 \end{bmatrix} \qquad v_2 = \begin{bmatrix} -1 \\ 6 \\ 5 \end{bmatrix} \qquad v_3 = \begin{bmatrix} 2 \\ 9 \\ -3 \end{bmatrix}$$

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 9 \\ -7 & 5 & -3 \end{bmatrix}$$

$$Ax = 0$$

Null Space & Linear Independence

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 9 \\ -7 & 5 & -3 \end{bmatrix}$$

$$Ax = 0$$

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ -2 & 6 & 9 & 0 \\ -7 & 5 & -3 & 0 \end{array}\right]$$

Augmented Matrix

Augmented Matrix

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ -2 & 6 & 9 & 0 \\ -7 & 5 & -3 & 0 \end{array}\right]$$

$R_3 \leftarrow 5R_3 + 7R_1$

$R_2 \leftarrow 5R_2 + 2R_1$

$R_2 \leftarrow \frac{R_2}{7}$

$R_3 \leftarrow 4R_3 - 18R_2$

$R_3 \leftarrow \frac{R_3}{10}$

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ 0 & 4 & 7 & 0 \\ 0 & 0 & -13 & 0 \end{array}\right]$$

Null Space & Linear Independence

Augmented Matrix

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ -2 & 6 & 9 & 0 \\ -7 & 5 & -3 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ 0 & 4 & 7 & 0 \\ 0 & 0 & -13 & 0 \end{array}\right]$$

$$\boxed{-13x_3 = 0}$$

$$x_1 = 0$$

$$x_2 = 0$$

$$x_3 = 0$$

$$\boxed{x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$$

Null Space & Linear Independence

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 9 \\ -7 & 5 & -3 \end{bmatrix} \qquad x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Ax = 0$$

$$v_1 = \begin{bmatrix} 5 \\ -2 \\ -7 \end{bmatrix} \qquad v_2 = \begin{bmatrix} -1 \\ 6 \\ 5 \end{bmatrix} \qquad v_3 = \begin{bmatrix} 2 \\ 9 \\ -3 \end{bmatrix}$$

Linearly Independent

Null Space & Linear Independence

$$v_1 = \begin{bmatrix} 5 \\ -2 \\ -7 \end{bmatrix} \qquad v_2 = \begin{bmatrix} -1 \\ 6 \\ 5 \end{bmatrix} \qquad v_3 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 2 \\ -7 & 5 & -1 \end{bmatrix}$$

$$Ax = 0$$

$$\rightarrow$$

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ -2 & 6 & 2 & 0 \\ -7 & 5 & -1 & 0 \end{array}\right]$$

Augmented Matrix

Null Space & Linear Independence

Augmented Matrix

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ -2 & 6 & 2 & 0 \\ -7 & 5 & -1 & 0 \end{array}\right]$$

$R_3 \leftarrow 5R_3 + 7R_1$

$R_3 \leftarrow \frac{R_2}{9}$

$R_2 \leftarrow 5R_2 + 2R_1$

$R_2 \leftarrow \frac{R_2}{14}$

$R_3 \leftarrow R_3 - R_2$

$$\longrightarrow \left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ -2 & 6 & 2 & 0 \\ -7 & 5 & -1 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 5 & -1 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\underline{2x_2 + x_3 = 0}$$

$$\underline{5x_1 - x_2 + 2x_3 = 0}$$

$$\boxed{x_3 \rightarrow free\ variable}$$

Null Space & Linear Independence

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 9 \\ -7 & 5 & -3 \end{bmatrix}$$

$$2x_2 + x_3 = 0$$

$$5x_1 - x_2 + 2x_3 = 0$$

$$Ax = 0$$

$$x_3 = 2$$

$$2x_2 + 2 = 0$$

$$2x_2 = -2$$

$$x_2 = -1$$

$$5x_1 - (-1) + 2 * 2 = 0$$

$$5x_1 + 1 + 4 = 0$$

$$5x_1 + 5 = 0$$

$$5x_1 = -5$$

$$x_1 = -1$$

$$x = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

Null Space & Linear Independence

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 2 \\ -7 & 5 & -1 \end{bmatrix} \qquad x = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$Ax = 0$$

$$v_1 = \begin{bmatrix} 5 \\ -2 \\ -7 \end{bmatrix} \qquad v_2 = \begin{bmatrix} -1 \\ 6 \\ 5 \end{bmatrix} \qquad v_3 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$$

Linearly Dependent

Column Space

- Column space of a matrix ***A*** (***col(A)***) is a set of all possible linear combinations of the column of the matrix.

$$Col(A) = span\{\, columns\ of\ Matrix\ A\}$$

$$\begin{aligned} 3x_1 - x_2 &= 0 \\ -6x_1 + 2x_2 &= 0 \end{aligned} \longrightarrow A = \begin{bmatrix} 3 & -1 \\ -6 & 2 \end{bmatrix}$$

$$Col(A) = span\left\{\begin{bmatrix} 3 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}\right\}$$

Column Space

- Column space of a matrix ***A*** (***col(A)***) is a set of all possible linear combinations of the column of the matrix.

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 2 \\ -7 & 5 & -1 \end{bmatrix}$$

$$Col(A) = span\left\{ \begin{bmatrix} 5 \\ -2 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 6 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \right\}$$

Column Space

Example

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 9 \\ -7 & 5 & -3 \end{bmatrix} \qquad b = \begin{bmatrix} 8 \\ 15 \\ 7 \end{bmatrix}$$

$$Ax = b$$

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 8 \\ -2 & 6 & 9 & 15 \\ -7 & 5 & -3 & 7 \end{array}\right]$$

Augmented Matrix

Column Space

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 9 \\ -7 & 5 & -3 \end{bmatrix} \qquad b = \begin{bmatrix} 8 \\ 15 \\ 7 \end{bmatrix}$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 5 & -1 & 2 & 8 \\ -2 & 6 & 9 & 15 \\ -7 & 5 & -3 & 7 \end{array}\right] \xrightarrow{\text{Row Reduced Echelon Form}} \begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} \\ \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 \end{array}\right] \end{array} \qquad \begin{array}{l} x_1 = 3 \\ x_2 = 5 \\ x_3 = -1 \end{array}$$

Rank & Nullity

- Dimension of a vector space is the number of basis vectors in the vector space.

- Dimension of a null space (Nullity) is the number of free variables or columns (columns that do not contain a pivot).

- Dimension of a column space (Rank) is the number of pivot columns.

Rank & Nullity

- To compute rank and nullity:
  1. Convert the matrix to RREF
  2. After row reduction, look for the first non-zero entry (pivot) in each row. The columns containing these pivot positions are called pivot columns.
  3. Any columns that do not contain pivot positions are free columns.

- Pivot columns form a basis for the column space of the matrix, and the number of pivot columns gives the rank of the matrix.

- Free columns indicate directions in the null space of the matrix, as they correspond to variables that can take arbitrary values without affecting the consistency of the system of equations.

Rank & Nullity

Example

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 2 \\ -7 & 5 & -1 \end{bmatrix} \xrightarrow{\text{Reduced Row Echelon Form}} \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix} \\ \begin{array}{cc} \textit{Pivot} & \textit{Free} \end{array} \end{array}$$

$Nullity: 1$

$Rank: 2$

Rank & Nullity

Example

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 9 \\ -7 & 5 & -3 \end{bmatrix} \xrightarrow{\textit{Reduced Row Echelon Form}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Pivot

$$Nullity: 0$$

$$Rank: 3$$

Rank-Nullity Theorem

$$A = \begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 2 \\ -7 & 5 & -1 \end{bmatrix} \xrightarrow{\text{Reduced Row Echelon Form}} \begin{array}{c} \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix} \\ \underbrace{\quad\quad}_{Pivot} \;\; \underbrace{\quad}_{Free} \end{array}$$

Rank: **number of pivot columns** — $Rank: 2$

Nullity: **number of free columns** — $Nullity: 1$

$$\boxed{Rank(A) + Nullity(A) = n} \longleftrightarrow$$

n is the number of columns in A

Orthogonal Vectors

- Vectors ***u*** and ***v*** are said to be **orthogonal** if **u.v = 0** or **u^{T}v = 0**.

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \qquad v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

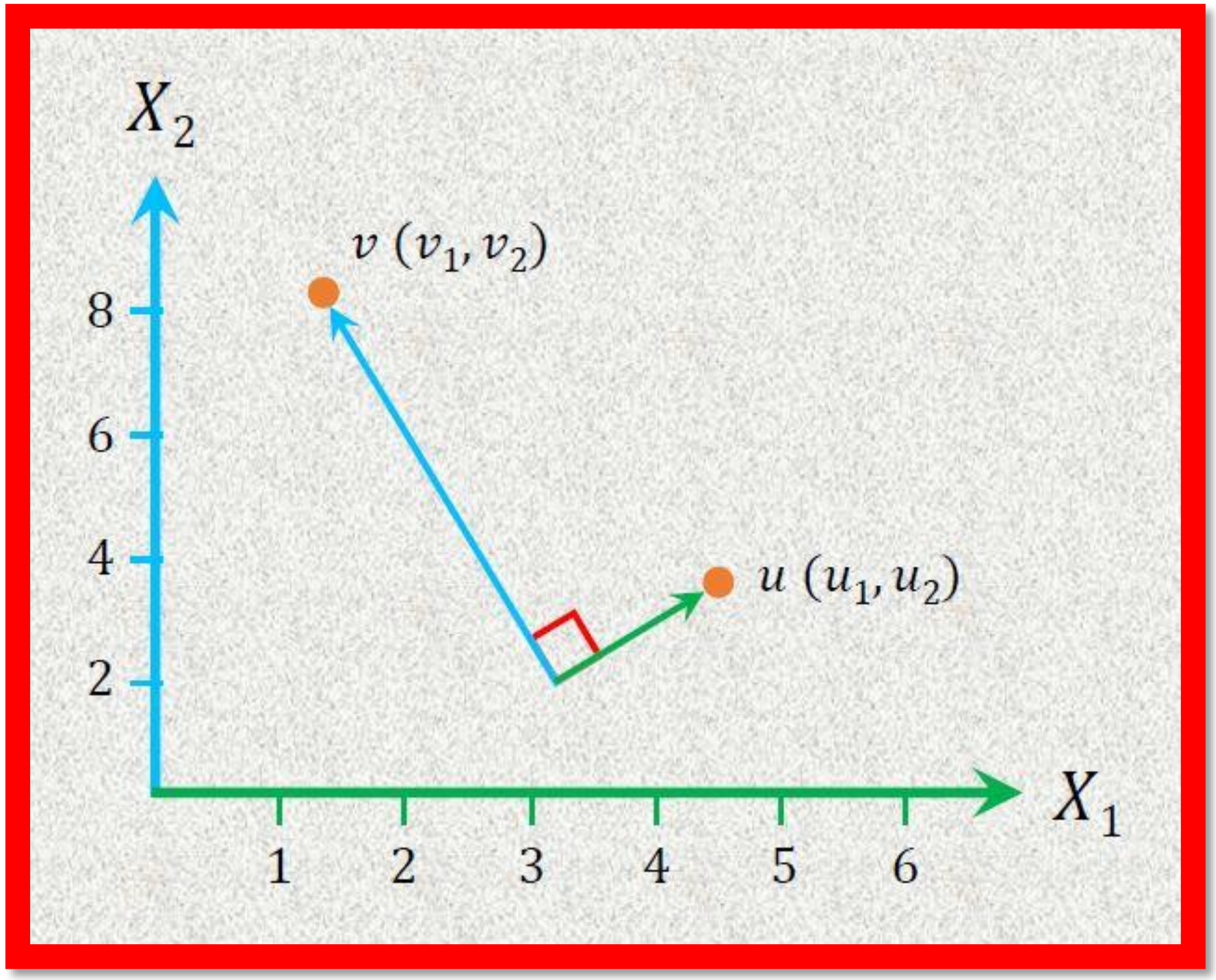
$$u.v = 0 \qquad u^T v = 0$$

$$\sum_{i=1}^{n} u_i \, v_i = 0 \qquad \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

Orthogonal Vectors

- Vectors ***u*** and ***v*** are said to be **orthogonal** if **u.v = 0** or **u^{T}v = 0**.

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \qquad v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$



Orthogonal Vectors

Example

$$u = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \qquad v = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$$

$$u.v = u^T v \quad = [\,1 \quad -2 \quad 4\,] \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$$

$$= 1 * 2 + (-2 * 5) + 4 * 2$$

$$= 2 - 10 + 8 \; = 0$$

Vectors u and v are Orthogonal Vectors

Orthogonal Vectors

Example

$$u = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \qquad v = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

$$u.v = u^T v = [1 \quad -2 \quad 4] \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

$$= 1 * 2 + (-2 * 5) + 4 * 1$$

$$= 2 - 10 + 4 = -4$$

Vectors u and v are not Orthogonal Vectors

Orthonormal Vectors

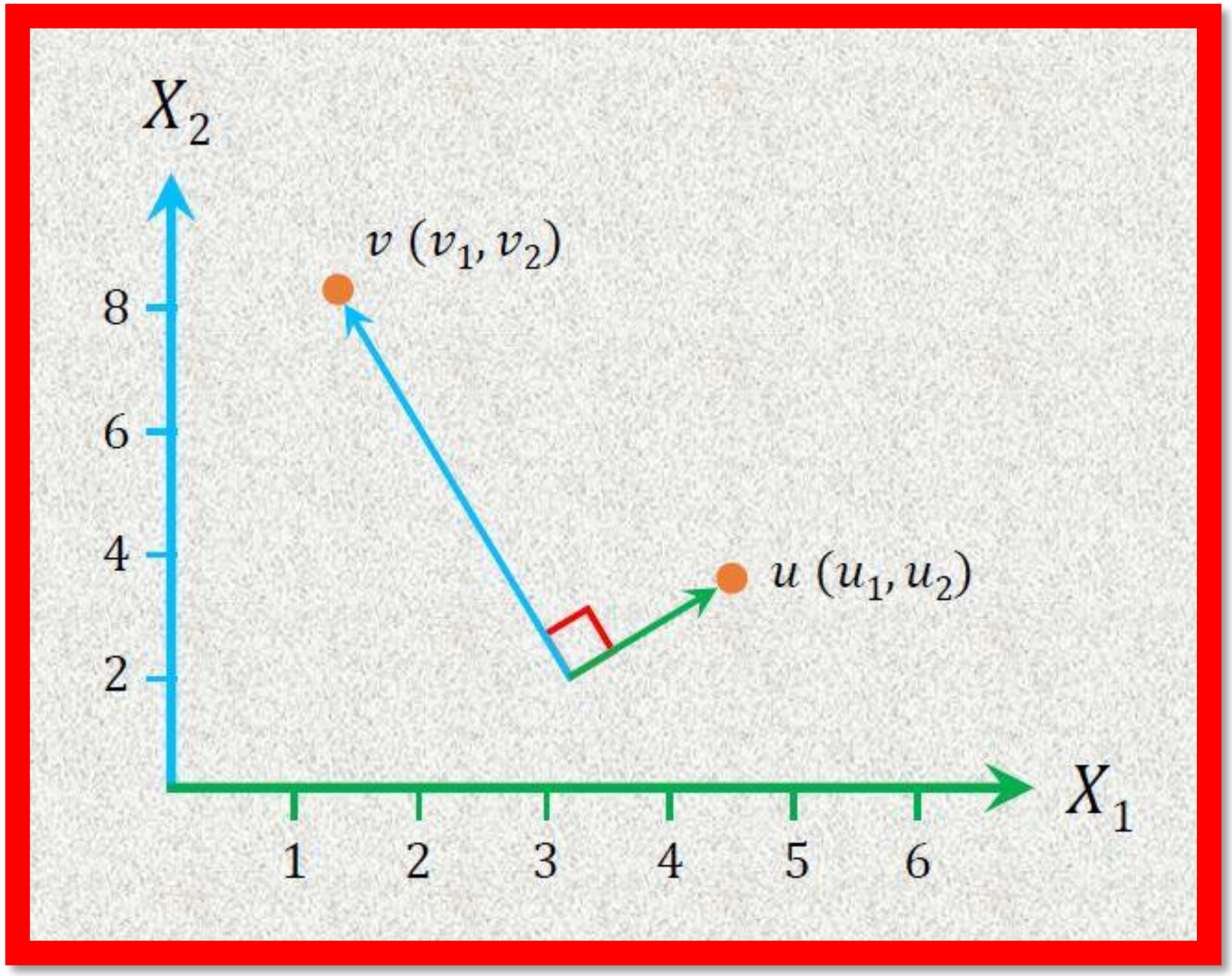
- Orthogonal vectors with <u>unit magnitude</u> is known as **Orthonormal** vectors.
- Orthonormal vectors are a set of vectors that are both <u>orthogonal</u> (perpendicular) to each other and <u>normalized</u> (each vector has unit length).

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \qquad v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\hat{u} = \frac{1}{|u|}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \qquad \hat{v} = \frac{1}{|v|}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\hat{u}.\,\hat{v} = 0 \qquad \hat{u}^T\hat{v} = 0$$

$$\sum_{i=1}^{n} \hat{u}_i\,\hat{v}_i = 0$$



Orthonormal Vectors

Example

$$u = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \quad v = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix} \qquad \hat{u} = \frac{1}{\sqrt{21}} \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \quad \hat{v} = \frac{1}{\sqrt{33}} \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$$

$$\hat{u}.\hat{v} = \hat{u}^T \hat{v} = \begin{bmatrix} \frac{1}{\sqrt{21}} & \frac{-2}{\sqrt{21}} & \frac{4}{\sqrt{21}} \end{bmatrix} \begin{bmatrix} 2/\sqrt{33} \\ 5/\sqrt{33} \\ 2/\sqrt{33} \end{bmatrix}$$

Orthonormal Vectors

$$= \frac{1}{\sqrt{21}} * \frac{2}{\sqrt{33}} + \frac{-2}{\sqrt{21}} * \frac{5}{\sqrt{33}} + \frac{4}{\sqrt{21}} * \frac{2}{\sqrt{33}}$$

$$= \frac{2}{\sqrt{21} * \sqrt{33}} - \frac{10}{\sqrt{21}\sqrt{33}} + \frac{8}{\sqrt{21}\sqrt{33}} = 0$$

Orthogonal Matrix

- Orthogonal Matrix: All the columns (and rows) of a square matrix are orthogonal to each other, and each column (and row) is a unit vector (has a length of one) – (both the columns and the rows of the matrix must be orthonormal to each other.)

the matrix must be an invertible matrix

$$M = \begin{bmatrix} m_{1,1} & m_{1,2} & . & . & m_{1,n} \\ m_{2,1} & m_{2,2} & . & . & . \\ . & . & . & . & . \\ . & . & . & . & . \\ m_{n,1} & m_{n,2} & . & . & m_{n,n} \end{bmatrix}$$

$$M^{-1} = M^T$$

$$M^{-1}M = I \qquad M^T M = I \qquad M M^T = I$$

Orthogonal Matrix

Example

$$M = \begin{bmatrix} 1/\sqrt{11} & 2/\sqrt{6} & -4/\sqrt{66} \\ 3/\sqrt{11} & -1/\sqrt{6} & -1/\sqrt{66} \\ 1/\sqrt{11} & 1/\sqrt{6} & 7/\sqrt{66} \end{bmatrix}$$

$$M^{-1} = M^T \qquad M^T M = I$$

$$\begin{bmatrix} 1/\sqrt{11} & 3/\sqrt{11} & 1/\sqrt{11} \\ 2/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \\ -4/\sqrt{66} & -1/\sqrt{66} & 7/\sqrt{66} \end{bmatrix} \begin{bmatrix} 1/\sqrt{11} & 2/\sqrt{6} & -4/\sqrt{66} \\ 3/\sqrt{11} & -1/\sqrt{6} & -1/\sqrt{66} \\ 1/\sqrt{11} & 1/\sqrt{6} & 7/\sqrt{66} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{\sqrt{11}} * \frac{1}{\sqrt{11}} + \frac{3}{\sqrt{11}} * \frac{3}{\sqrt{11}} + \frac{1}{\sqrt{11}} * \frac{1}{\sqrt{11}} & \frac{1}{\sqrt{11}} * \frac{2}{\sqrt{6}} + \frac{3}{\sqrt{11}} * \frac{-1}{\sqrt{6}} + \frac{1}{\sqrt{11}} * \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{11}} * \frac{-4}{\sqrt{66}} + \frac{3}{\sqrt{11}} * \frac{-1}{\sqrt{66}} + \frac{1}{\sqrt{11}} * \frac{7}{\sqrt{66}} \\ \frac{2}{\sqrt{6}} * \frac{1}{\sqrt{11}} + \frac{-1}{\sqrt{6}} * \frac{3}{\sqrt{11}} + \frac{1}{\sqrt{6}} * \frac{1}{\sqrt{11}} & \frac{2}{\sqrt{6}} * \frac{2}{\sqrt{6}} + \frac{-1}{\sqrt{6}} * \frac{-1}{\sqrt{6}} + \frac{1}{\sqrt{6}} * \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} * \frac{-4}{\sqrt{66}} + \frac{-1}{\sqrt{6}} * \frac{-1}{\sqrt{66}} + \frac{1}{\sqrt{6}} * \frac{7}{\sqrt{66}} \\ \frac{-4}{\sqrt{66}} * \frac{1}{\sqrt{11}} + \frac{-1}{\sqrt{66}} * \frac{3}{\sqrt{11}} + \frac{7}{\sqrt{66}} * \frac{1}{\sqrt{11}} & \frac{-4}{\sqrt{66}} * \frac{2}{\sqrt{6}} + \frac{-1}{\sqrt{66}} * \frac{-1}{\sqrt{6}} + \frac{7}{\sqrt{66}} * \frac{1}{\sqrt{6}} & \frac{-4}{\sqrt{66}} * \frac{-4}{\sqrt{66}} + \frac{-1}{\sqrt{66}} * \frac{-1}{\sqrt{66}} + \frac{7}{\sqrt{66}} * \frac{7}{\sqrt{66}} \end{bmatrix}$$

Orthogonal Matrix

Example

$$\begin{bmatrix} \frac{1}{11}+\frac{9}{11}+\frac{1}{11} & \frac{2}{\sqrt{11}\sqrt{6}}-\frac{3}{\sqrt{11}\sqrt{6}}+\frac{1}{\sqrt{11}\sqrt{6}} & \frac{-4}{\sqrt{11}\sqrt{66}}-\frac{3}{\sqrt{11}\sqrt{66}}+\frac{7}{\sqrt{11}\sqrt{66}} \\ \frac{2}{\sqrt{6}\sqrt{11}}+\frac{-3}{\sqrt{6}\sqrt{11}}+\frac{1}{\sqrt{6}\sqrt{11}} & \frac{4}{6}+\frac{1}{6}+\frac{1}{6} & \frac{-8}{\sqrt{6}\sqrt{66}}+\frac{1}{\sqrt{6}\sqrt{66}}+\frac{7}{\sqrt{6}\sqrt{66}} \\ \frac{-4}{\sqrt{66}\sqrt{11}}+\frac{-3}{\sqrt{66}\sqrt{11}}+\frac{7}{\sqrt{66}\sqrt{11}} & \frac{-8}{\sqrt{66}\sqrt{6}}+\frac{1}{\sqrt{66}\sqrt{6}}+\frac{7}{\sqrt{66}\sqrt{6}} & \frac{16}{66}+\frac{1}{66}+\frac{49}{66} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1+9+1}{11} & \frac{2-3+1}{\sqrt{11}\sqrt{6}} & \frac{-4-3+7}{\sqrt{11}\sqrt{66}} \\ \frac{2-3+1}{\sqrt{6}\sqrt{11}} & \frac{4+1+1}{6} & \frac{-8+1+7}{\sqrt{6}\sqrt{66}} \\ \frac{-4-3+7}{\sqrt{66}\sqrt{11}} & \frac{-8+1+7}{\sqrt{66}\sqrt{6}} & \frac{16+1+49}{66} \end{bmatrix}$$

$$\begin{bmatrix} \frac{11}{11} & \frac{0}{\sqrt{11}\sqrt{6}} & \frac{0}{\sqrt{11}\sqrt{66}} \\ \frac{0}{\sqrt{6}\sqrt{11}} & \frac{6}{6} & \frac{0}{\sqrt{6}\sqrt{66}} \\ \frac{0}{\sqrt{66}\sqrt{11}} & \frac{0}{\sqrt{66}\sqrt{6}} & \frac{66}{66} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Orthogonal Matrix

Example

$$M = \begin{bmatrix} 1/\sqrt{11} & 2/\sqrt{6} & -4/\sqrt{66} \\ 3/\sqrt{11} & -1/\sqrt{6} & -1/\sqrt{66} \\ 1/\sqrt{11} & 1/\sqrt{6} & 7/\sqrt{66} \end{bmatrix}$$

$$M^{-1} = M^T \qquad M^T M = I$$

$$\begin{bmatrix} 1/\sqrt{11} & 3/\sqrt{11} & 1/\sqrt{11} \\ 2/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \\ -4/\sqrt{66} & -1/\sqrt{66} & 7/\sqrt{66} \end{bmatrix} \begin{bmatrix} 1/\sqrt{11} & 2/\sqrt{6} & -4/\sqrt{66} \\ 3/\sqrt{11} & -1/\sqrt{6} & -1/\sqrt{66} \\ 1/\sqrt{11} & 1/\sqrt{6} & 7/\sqrt{66} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore, the Matrix M is <u>Orthogonal Matrix</u>

Null Space

Exercise

Consider the following matrix *A*. Find the null space of *A*.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Null Space

Exercise

Consider the following matrix B. Find the null space of B.

$$B = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & -1 \\ 3 & -2 & 2 \end{bmatrix}$$

Column Space

Exercise

Consider the following matrix *A*. Find the column space of *A*.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Column Space

Exercise

Consider the following matrix *B*. Find the column space of *B*.

$$B = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 4 \\ 3 & 9 & 6 \end{bmatrix}$$

Rank & Nullity

Exercise

Consider the following matrix *A*. Determine the rank and nullity of *A*.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

End of Lecture 7